“Tanglaw: Bringing Trusted Information Within Reach, Anywhere”

Track: **“**Community Impact: MIL-based interventions that empower communities”

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1. Problem Statement

The Philippines Does Not Face a Misinformation Problem Alone, It Faces an Information Accessibility Problem

Every day, millions of Filipinos receive disaster warnings, financial messages, government advisories, and community updates through their mobile phones. Yet for many of the country's most vulnerable citizens, verifying whether these messages are trustworthy is often impossible at the moment when it matters most. During disasters, scams, and public emergencies, accurate information may already exist, but access to that information does not. When internet connectivity is unavailable, unaffordable, or disrupted, people are forced to make critical decisions based on incomplete, unverified, or misleading information.

This proposal argues that the challenge extends beyond misinformation itself. **Misinformation is not only a content problem, it is an information accessibility problem.** Communities cannot fully practice Media and Information Literacy (MIL) if they are unable to access, compare, evaluate, and verify information when they need it most. In low-connectivity environments, critical thinking is constrained not by willingness, but by infrastructure, affordability, and unequal access to trustworthy sources.

This challenge directly reflects the UNESCO Youth Hackathon 2026 theme, **"Play Your Part: Youth Designing the Future of Media and Information Literacy."** UNESCO defines MIL as empowering individuals to access, analyze, evaluate, create, and responsibly use information for informed decision-making, democratic participation, resilience, and lifelong learning (UNESCO, 2023; UNESCO, 2026a). For many digitally marginalized Filipino communities, however, these competencies remain difficult to exercise because reliable information is often inaccessible precisely when it is needed most.

Current Situation

The Philippines presents an especially urgent context for this challenge. As one of the world's most disaster-prone archipelagos, the country experiences frequent typhoons, floods, earthquakes, volcanic eruptions, and other emergencies that depend on timely and trustworthy public communication. At the same time, Filipinos are among the world's most active users of digital communication platforms, making online spaces critical channels for both official information and misinformation.

Despite continued improvements in digital infrastructure, meaningful internet access remains uneven. According to the Philippine Statistics Authority, only **48.8% of Filipino**

households had internet access at home in 2024 (PSA, 2026). Many households continue to rely on prepaid mobile data, shared devices, intermittent cellular connections, or public Wi-Fi. These challenges are particularly severe in geographically isolated and disadvantaged areas (GIDAs), island municipalities, and rural barangays where network coverage remains inconsistent (World Bank, 2024).

As a result, verifying information often carries an additional financial cost. While misinformation spreads rapidly through text messages, messaging applications, and social media, confirming those claims frequently requires accessing government advisories, independent fact-checking platforms, or trusted news websites that consume additional mobile data. Consequently, the people most vulnerable to misinformation are often those with the fewest opportunities to verify it.

These barriers are not isolated problems. Together, they reinforce one another and create a cycle of information inequality that existing MIL interventions struggle to address.

1. The "Walled Garden" Subsidy: How Zero-Rated Platforms Penalize Verification

While basic connectivity in the Philippines has expanded, it has done so through a highly uneven economic model. Mobile network operators frequently offer zero-rated "free basic" data packages that grant unlimited access to proprietary social media feeds and messaging applications but restrict access to the broader web. This creates a structural "walled garden." Within this digital ecosystem, consuming and forwarding unverified claims, sensationalized headlines, and fabricated emergency alerts costs a user nothing.

Conversely, verifying those claims, by clicking external links to read official government bulletins, cross-referencing independent fact-checking databases, or accessing trusted media outlets requires a paid mobile data balance. Consequently, data poverty imposes an active financial penalty on media literacy. The poorest, most vulnerable segments of the population are systematically priced out of the ability to practice critical evaluation, transforming Media and Information Literacy (MIL) from an educational habit into an economic privilege.

2. Disasters Intensify Information Vulnerability

Information inequality becomes especially dangerous during disasters. Typhoons, volcanic eruptions, floods, earthquakes, and other emergencies require communities to make rapid decisions based on accurate information. However, these same situations also create ideal conditions for misinformation, manipulated media, and fabricated emergency announcements to spread through informal communication networks.

UNESCO emphasizes that crisis environments significantly increase the importance of trustworthy information because communities must make high-risk decisions under uncertainty and disrupted communication systems (UNESCO, 2026b). Philippine experiences during events such as Typhoon Haiyan, Typhoon Odette, Typhoon Egay, and the Taal Volcano eruption demonstrate how false evacuation notices, misleading videos, and fabricated advisories circulated alongside legitimate government warnings (VERA Files, 2020a; 2020b; 2022; 2025).

The challenge is therefore not simply whether official information exists, but whether affected communities can access and verify it before misinformation influences their decisions.

3. Limited Verification Increases Exposure to Digital Fraud

The same verification gap also increases communities' vulnerability to scams and social engineering.

Cybercriminals increasingly exploit trusted communication channels through phishing links, impersonation scams, fraudulent financial messages, and fake government assistance announcements. Recent reports associated with the Bangko Sentral ng Pilipinas indicate that cyber-related financial losses continue to increase, while phishing and transaction fraud remain major consumer protection concerns (2025a; 2025b).

These attacks succeed because they exploit urgency, trust, and limited opportunities for verification. For senior citizens, informal workers, students, overseas Filipino worker (OFW) families, and low-income households, verifying suspicious messages often requires additional mobile data or navigating multiple websites. When verification is difficult, immediate action frequently replaces informed decision-making.

4. Existing Solutions Assume Reliable Internet Access

Although fact-checking organizations, digital literacy campaigns, and online verification tools continue to expand, most existing interventions assume that users have reliable internet access whenever verification is needed.

This assumption does not consistently reflect the realities of many Filipino communities.

Most verification platforms require users to search online, open external links, access cloud-based databases, or stream educational content. While these approaches remain valuable, they are far less effective for individuals making urgent decisions in disaster zones, remote barangays, informal settlements, or low-income communities where connectivity is intermittent, expensive, or unavailable.

As a result, the communities that would benefit most from Media and Information Literacy interventions are often those least able to access them.

Together, these structural factors create a persistent cycle of information inequality: limited connectivity reduces access to trustworthy information, weaker verification increases vulnerability to misinformation and fraud, and repeated exposure ultimately erodes community resilience, public trust, and informed civic participation.

Breaking this cycle requires more than another online fact-checking platform. It requires reimagining Media and Information Literacy itself, designing solutions that work within the realities of digitally marginalized communities rather than assuming universal connectivity.

2. Solution Description

Overview

Tanglaw P2P is an **offline-first Community Verification Infrastructure** that enables communities to practice Media and Information Literacy even when internet connectivity is limited, expensive, or unavailable.

Rather than treating misinformation as only a content moderation problem, Tanglaw P2P addresses its underlying structural cause: **unequal access to trustworthy information.**

The platform enables users to verify suspicious messages, recognize manipulation techniques, and make informed decisions without depending on continuous internet access. By combining offline verification, lightweight synchronization, community knowledge sharing, and accessible design, Tanglaw P2P transforms Media and Information Literacy from an online activity into an everyday community capability.

Instead of asking vulnerable communities to adapt to digital systems designed for highly connected environments, the platform adapts digital verification to the realities of disaster-prone, low-bandwidth, and digitally marginalized communities.

In doing so, Tanglaw P2P directly advances UNESCO's vision of Media and Information Literacy by helping individuals **access, evaluate, understand, and responsibly share information** regardless of connectivity constraints.

The Core Innovation

Most misinformation solutions begin with a simple assumption:

> Users can connect to the internet whenever they need to verify information.

Tanglaw P2P challenges that assumption.

The platform introduces a fundamentally different model:

> **Verification should travel with the community, not remain dependent on the internet.**

Instead of requiring users to search online every time they receive suspicious information, Tanglaw P2P places essential verification resources directly on users' devices through an offline-first architecture that continues functioning during connectivity disruptions, disasters, and periods of limited mobile data.

This shift fundamentally changes how Media and Information Literacy is practiced.

Rather than treating verification as a web service, the project treats verification as **community infrastructure**, a shared capability that remains available whenever people need trustworthy information most.

Three Innovation Pillars

Instead of functioning as a traditional fact-checking application, Tanglaw P2P is designed around three interconnected pillars that collectively strengthen community information resilience.

Pillar 1 — Accessible Verification Anywhere

The first pillar ensures that trustworthy information remains accessible regardless of internet availability.

At the heart of the platform is the **Offline Threat Ledger**, a lightweight, locally stored repository containing verified scam patterns, common misinformation indicators, suspicious sender identifiers, emergency verification guidance, and locally relevant public information.

Because these resources are stored directly on the user's device, verification continues even during:

- limited mobile data,
- unstable cellular service,
- disaster-related network outages,
- power interruptions,
- or remote community deployment.

Instead of opening multiple websites or consuming expensive mobile data, users receive immediate guidance using information already available on their devices.

This significantly reduces one of the largest barriers to Media and Information Literacy: the inability to access trustworthy information when connectivity fails.

Pillar 2 — Intelligent Community Verification

Access alone is not sufficient.

Communities must also understand **why** information may be trustworthy, or deceptive.

To support this, Tanglaw P2P combines lightweight local analysis with community-driven verification.

The platform includes a **Localized Text Parser** that examines suspicious text messages, links, and short claims directly on the user's device without transmitting personal data to external servers.

Rather than functioning as a black-box AI system, the parser explains the characteristics that contribute to potential risk, including:

- impersonation of trusted institutions,

- artificial urgency,
- suspicious links,
- financial pressure,
- inconsistent language,
- commonly reported scam patterns,
- and other indicators of manipulation.

Instead of simply telling users that a message is "fake," the platform teaches users **how to recognize deceptive information themselves.**

This transforms verification into an educational experience that continuously strengthens Media and Information Literacy rather than merely providing isolated answers.

Whenever connectivity becomes available, users receive compact updates to the Threat Ledger through lightweight synchronization while maintaining local privacy and minimizing bandwidth consumption.

Pillar 3 — Decentralized Community Replication & Peer-to-Peer Sync

Information in Filipino communities is fundamentally relational, spreading organically through local hubs, neighbors, and family networks. Tanglaw P2P mirrors this behavior through a decentralized, offline distribution architecture. Instead of relying on a centralized server that becomes inaccessible during disasters or network congestion, the platform distributes the **Offline Threat Ledger** via dynamic local sync mechanisms:

Dynamic Local Sync: When a "seed user" (such as a local student advocate or barangay health worker) connects to the internet, their device automatically pulls a highly compressed ledger update (<1 MB).

Local P2P Distribution: Using WebRTC for browser-to-browser data channels or localized Wi-Fi Direct, the seed user can instantly replicate the updated threat signatures to offline devices in their immediate physical proximity, requiring zero cellular data or internet access.

QR-Code Fallback: For ultra-low-end legacy smartphones, the app can render a high-density, dynamic QR code containing the latest critical threat hashes. Nearby users simply scan the QR code within their Tanglaw app to import the offline update instantly.

This approach transforms media verification from a series of isolated internet queries into a shared, self-sustaining community utility that grows more resilient as more local nodes join the network.

Designed for Inclusive Media and Information Literacy

Digital inclusion requires more than internet connectivity.

It also requires interfaces that accommodate different literacy levels, languages, ages, and abilities.

Tanglaw P2P therefore adopts a **Universal Access Design** that ensures verification remains usable across diverse communities.

The platform incorporates:

- voice-assisted guidance,
- simplified language,
- multilingual support,
- intuitive visual indicators,
- high-contrast accessibility,
- and low-bandwidth interaction patterns.

Instead of overwhelming users with technical explanations, information is presented through clear risk categories, understandable language, and actionable recommendations.

This enables senior citizens, students, informal workers, persons with disabilities, and first-time smartphone users to participate confidently in Media and Information Literacy regardless of previous digital experience.

Accessibility is therefore treated not as an additional feature, but as a fundamental design principle.

Supporting Communities During Crises

Information becomes most valuable when communities are under pressure.

During typhoons, floods, volcanic eruptions, earthquakes, conflict displacement, or emerging scam campaigns, Tanglaw P2P automatically prioritizes time-sensitive verification through **Crisis Verification Mode**.

Rather than overwhelming users with large volumes of information, the platform highlights urgent messages, recommends trusted verification pathways, and presents concise guidance that supports rapid decision-making.

The goal is not to replace official government advisories.

Instead, the platform helps communities interpret those advisories more confidently while reducing the influence of rumors, manipulated content, and fraudulent emergency messages.

By strengthening verification precisely when misinformation causes the greatest harm, the platform contributes directly to community resilience.

A Community-Centered User Journey

The platform has been designed around the everyday experience of its users rather than around technology itself.

Consider a typical scenario.

A resident receives a forwarded text message claiming that a nearby evacuation center has been relocated because of an approaching typhoon.

Without sufficient mobile data, visiting multiple government websites or searching online becomes difficult.

Instead, the resident opens Tanglaw P2P and pastes the message into the application.

Within seconds, the Localized Text Parser analyzes the content locally, compares it against the Offline Threat Ledger, and explains the indicators that suggest whether the message appears trustworthy, suspicious, or requires further verification.

The application recommends official sources where appropriate while clearly explaining its confidence level rather than making absolute claims.

Later, when internet connectivity becomes available, the user's device receives a lightweight update containing newly verified scam patterns and public advisories.

Those updates can then be shared efficiently throughout nearby households and community partners, ensuring that verification capacity spreads together with trusted information.

In this way, verification evolves from an individual action into a collective community practice.

Alignment with UNESCO Media and Information Literacy

Every component of Tanglaw P2P is intentionally designed to strengthen Media and Information Literacy rather than simply detect misinformation.

The platform enables communities to:

- **Access** trustworthy information despite connectivity limitations.
- **Evaluate** suspicious content using locally available verification tools.
- **Analyze** manipulation techniques through transparent explanations rather than opaque algorithms.
- **Create and Share Responsibly** by encouraging evidence-based community reporting.
- **Participate** more confidently in civic life through equitable access to trustworthy information.

Rather than functioning solely as a cybersecurity application, Tanglaw P2P becomes a practical, everyday Media and Information Literacy intervention that empowers digitally marginalized communities to make informed decisions, strengthen resilience, and participate more safely within the modern information ecosystem.

3. Implementation Plan

Implementation Philosophy

Tanglaw P2P will be implemented through a community-centered, iterative deployment strategy that prioritizes usability, accessibility, and local adoption before large-scale expansion. Rather than developing a fully featured platform in isolation, the project will evolve alongside its users through continuous testing, community feedback, and partnerships with local stakeholders.

This phased approach reflects UNESCO's principle that effective Media and Information Literacy (MIL) interventions should be developed with communities rather than merely for communities. Each implementation phase therefore combines technical development with user validation, accessibility improvements, and capacity building to ensure the platform remains practical in low-resource environments.

The implementation strategy is guided by four principles:

- Offline-first by design
- Community-driven by deployment
- Privacy-preserving by architecture

- Scalable by replication

These principles ensure that technological development always serves the broader goal of strengthening community information resilience.

Phase 1 — Establish the Community Verification Foundation

The first phase focuses on building the platform's core capability: enabling communities to verify information even without continuous internet connectivity.

Development begins with the Progressive Web Application (PWA), secure local device storage, and the initial Offline Threat Ledger containing verified scam indicators, misinformation patterns, and emergency verification resources. These components establish the minimum functionality required for users to evaluate suspicious information regardless of network availability.

Rather than prioritizing advanced features, this phase ensures that the platform solves the project's central challenge from the beginning: **verification must remain available even when connectivity does not.**

Expected Outcome

- Functional offline verification prototype
- Initial Threat Ledger
- Secure local storage architecture
- Core user verification workflow

Phase 2 — Build Intelligent Verification and Accessibility

Once offline functionality has been established, the second phase strengthens the quality and inclusiveness of the verification experience.

This phase introduces the Localized Text Parser capable of recognizing common characteristics of scams, phishing attempts, manipulated emergency messages, and misleading claims frequently encountered within the Philippine digital environment.

Unlike conventional automated detection systems, the parser provides transparent explanations that help users understand *why* content may be suspicious. This transforms every verification into a Media and Information Literacy learning opportunity rather than simply producing a risk label.

Accessibility enhancements, including multilingual support, voice guidance, simplified interfaces, and intuitive visual indicators, are implemented simultaneously to ensure that the platform remains usable across different literacy levels, age groups, and accessibility needs.

Expected Outcome

- Explainable verification engine
- Voice-first interface
- Multilingual accessibility
- Inclusive user experience for diverse communities

Phase 3 — Strengthen Community Information Resilience

The third phase expands the platform beyond individual users by introducing community-based information sharing.

Hybrid Synchronization enables lightweight updates to the Offline Threat Ledger whenever internet access becomes available. Rather than requiring every user to download large datasets independently, compact verification updates can be efficiently distributed across participating devices while minimizing bandwidth consumption.

Community Replication mechanisms allow trusted community members, households, and local hubs to share updated verification resources through low-cost transfer methods, ensuring that trustworthy information continues to circulate even where connectivity remains intermittent.

This phase recognizes that information resilience is fundamentally collective.

Communities become stronger when trusted information spreads as efficiently as misinformation.

Expected Outcome

- Lightweight synchronization system
- Community replication workflow
- Shared Threat Ledger updates
- Increased resilience during connectivity disruptions

Phase 4 — Optimize for Crisis Response

The fourth phase prepares the platform for real-world emergencies where timely information becomes most critical.

Crisis Verification Mode is refined to prioritize disaster-related messages, emergency advisories, and rapidly emerging misinformation events. Instead of overwhelming users with technical details, the platform delivers concise risk explanations, trusted verification pathways, and context-specific guidance that supports rapid decision-making under pressure.

The interface is optimized for stressful situations where users may have limited time, limited connectivity, or heightened emotional vulnerability.

By supporting informed decisions during disasters, the platform directly contributes to safer communities and stronger public resilience.

Expected Outcome

- Crisis Verification Mode
- Emergency-focused user interface
- Rapid verification workflows
- Improved disaster information resilience

Phase 5 — Pilot with Communities

Technology alone cannot determine whether a Media and Information Literacy intervention succeeds.

The fifth phase therefore focuses on participatory pilot implementation within selected barangays, educational institutions, and community organizations representing diverse connectivity conditions.

Rather than measuring only technical performance, pilot testing evaluates:

- usability,
- community trust,
- accessibility,
- verification behavior,
- adoption patterns,
- and changes in Media and Information Literacy practices.

Community members become active co-designers by providing feedback on interface design, local language, verification explanations, and overall usability.

This participatory approach ensures that the platform reflects real community needs instead of assumptions made during development.

Potential Pilot Partners

- Barangay Local Government Units
- Public schools and universities
- Youth organizations
- Community disaster response volunteers
- Civil society organizations
- Digital literacy advocates

Expected Outcome

- Community-validated platform
- User-centered design improvements
- Increased trust and adoption
- Locally informed feature refinement

Phase 6 — Scale Through Partnerships

Following successful pilot validation, the final phase focuses on sustainable expansion through institutional partnerships rather than centralized deployment.

Instead of relying on a single organization to maintain nationwide operations, Tanglaw P2P is designed as a replicable community verification model that can be adopted by local governments, educational institutions, civil society organizations, humanitarian agencies, and community networks.

Scaling efforts prioritize knowledge transfer, documentation, and training materials so that communities can independently deploy and sustain the platform with minimal technical barriers.

Potential collaboration opportunities include organizations working in disaster preparedness, digital inclusion, cybersecurity awareness, and Media and Information Literacy promotion.

By emphasizing community ownership over centralized control, the project strengthens long-term sustainability while supporting broader national adoption.

Expected Outcome

- Partnership framework
- Deployment toolkit

- Community training resources
- Scalable implementation model
- Sustainable governance pathway

Monitoring, Evaluation, and Continuous Improvement

Implementation does not conclude after deployment.

Throughout every phase, the project follows a continuous improvement cycle that combines technical evaluation with community feedback.

Success will be monitored using both technical and social impact indicators, including:

Technical Indicators

- Offline verification success rate
- Synchronization reliability
- System availability
- Low-bandwidth performance
- Accessibility compliance

Community Indicators

- Adoption and retention rates
- Community trust in verification results
- Improvement in Media and Information Literacy competencies
- User confidence in identifying misinformation
- Reduction in reported scam susceptibility
- Satisfaction with accessibility features

Insights gathered during each phase will inform iterative improvements, ensuring that the platform continues evolving alongside the needs of its users.

Long-Term Sustainability

From the outset, Tanglaw P2P is designed not as a one-time hackathon prototype, but as a sustainable community resource.

Long-term sustainability will be supported through four complementary strategies:

- **Community Ownership:** Empower local leaders, educators, and volunteers to promote and sustain platform use.
- **Institutional Partnerships:** Collaborate with government agencies, educational institutions, NGOs, and Media and Information Literacy advocates to integrate the platform into existing community programs.
- **Open and Modular Architecture:** Maintain a lightweight, extensible system that can be adapted to different local contexts without extensive redevelopment.
- **Continuous Learning:** Regularly update verification resources and educational content based on emerging threats, community feedback, and evolving misinformation patterns.

This approach enables the platform to remain relevant, maintainable, and responsive long after the initial pilot phase.

Implementation Timeline

Phase	**Primary Objective**	**Community Outcome**
Phase 1	Build Offline Verification Foundation	Communities can verify information without continuous internet access.
Phase 2	Introduce Explainable Verification and Accessibility	Users understand *why* information may be misleading while ensuring inclusivity.
Phase 3	Enable Community Replication	Trusted verification resources spread efficiently throughout communities.
Phase 4	Optimize for Crisis Response	Communities make safer decisions during disasters and emergencies.
Phase 5	Community Pilot and Validation	The platform is refined through real-world feedback and local participation.
Phase 6	Scale Through Partnerships	Sustainable expansion through community ownership and institutional collaboration.

4. Target Audience and User Personas

Who We Serve

Tanglaw P2P is designed for communities where access to trustworthy information cannot be taken for granted. Rather than targeting the general smartphone population, the platform prioritizes **digitally marginalized individuals** whose ability to practice Media and Information Literacy (MIL) is constrained by limited connectivity, affordability, geography, disaster vulnerability, or accessibility barriers.

The project recognizes that misinformation does not affect all people equally. Communities with fewer opportunities to verify information are often the same communities that experience the greatest consequences when misinformation spreads, whether through financial loss, delayed disaster response, health misinformation, or reduced trust in public institutions.

For this reason, Tanglaw P2P follows an **equity-first approach**, focusing on users who face the greatest barriers to accessing and evaluating trustworthy information.

Primary Beneficiaries

1. Low-Income Households

Many Filipino families rely on prepaid mobile data, shared devices, or intermittent internet access. Because online verification often requires additional data, many households receive information through messaging applications without being able to confirm its accuracy immediately.

For these users, Tanglaw P2P provides offline verification tools that reduce the cost of practicing Media and Information Literacy while strengthening confidence in everyday decision-making.

2. Disaster-Prone Communities

Residents living in areas frequently affected by typhoons, floods, earthquakes, volcanic eruptions, or conflict-related displacement often make life-critical decisions under severe time pressure.

When communication infrastructure becomes disrupted, misinformation can spread alongside official emergency advisories.

Tanglaw P2P supports these communities by providing offline verification resources, crisis-specific guidance, and lightweight updates that remain accessible even when internet connectivity becomes unreliable.

3. Rural, Remote, and Geographically Isolated Communities (GIDAs)

Many geographically isolated and disadvantaged areas experience inconsistent connectivity despite increasing smartphone adoption.

These communities often receive public information through local leaders, radio broadcasts, text messaging, and interpersonal communication rather than continuous internet access.

By functioning offline and supporting community-based knowledge sharing, Tanglaw P2P complements existing local communication practices instead of replacing them.

4. Older Adults and Digitally Vulnerable Individuals

Senior citizens are disproportionately affected by phishing attacks, impersonation scams, fraudulent financial messages, and misinformation campaigns.

Many also experience challenges navigating complex digital interfaces.

Voice-assisted guidance, simplified language, visual indicators, and explainable verification enable these users to practice Media and Information Literacy with greater confidence while reducing dependence on technically skilled family members.

5. Students and Young People

Young people are among the most active consumers and sharers of digital information.

As future community leaders and peer educators, they play a critical role in strengthening responsible information practices.

Rather than functioning solely as end users, students become ambassadors for Media and Information Literacy by helping families and communities adopt responsible verification behaviors.

This aligns directly with UNESCO's vision of youth as active contributors to resilient information ecosystems.

6. Strict Mitigation and Ethical Guardrails

The "Neutral/Unknown State" Protocol: If a user parses a message during an active emergency that does not match any entry in the Offline Threat Ledger, the app must strictly avoid labeling it as "Safe" or "Fake". Labeling an unverified, real-time warning as false could delay critical actions, while labeling a new scam as safe could lead to harm. Instead, the system triggers a **"Neutral/Unknown"** status, displaying a standard offline emergency checklist (e.g., verifying with physical barangay captains).

Cryptographically Signed Ledger Integrity: To prevent malicious actors from distributing corrupted or politically biased "threat ledgers" to offline communities, all Ledger update packages are cryptographically signed by verified, trusted authorities (e.g., vetted fact-checkers, local disaster agencies). The local client app will verify the digital signature using a pre-installed public key before importing any offline updates.

Secondary Stakeholders

The platform also creates value for organizations that support community information resilience.

These include:

- Barangay Local Government Units (LGUs)
- Schools, universities, and educators
- Community disaster risk reduction offices
- Civil society organizations
- Youth organizations
- Media and Information Literacy advocates
- Community health workers
- Humanitarian organizations
- Financial literacy and cybersecurity initiatives

These stakeholders can use aggregated community insights, educational materials, and verified threat patterns to strengthen local awareness campaigns and improve community preparedness.

User Persona 1 — Maria

The Rural Mother

Age: 39

Location: Rural barangay in Quezon Province

Occupation: Small-scale farmer

Connectivity: Prepaid mobile data with intermittent signal

Situation

Maria receives a forwarded text message claiming that relief distribution has been moved because of an approaching typhoon.

Opening multiple government websites would consume the remaining balance on her prepaid mobile data.

She must decide quickly whether to travel to another barangay.

Current Challenge

Unable to verify the information immediately, Maria depends on neighbors and forwarded messages, increasing the likelihood of acting on inaccurate information.

How Tanglaw P2P Helps

Maria pastes the message into the application.

The Offline Threat Ledger recognizes similar misinformation patterns previously reported during disaster responses.

The app explains why the message appears suspicious, recommends trusted verification sources, and provides guidance without requiring continuous internet access.

Instead of reacting to rumors, Maria makes a more informed decision based on available evidence.

User Persona 2 — Roberto

The Senior Citizen

Age: 67

Location: Batangas

Occupation: Retired

Connectivity: Smartphone with limited digital confidence

Situation

Roberto receives a text message claiming that his bank account will be suspended unless he immediately confirms personal information.

The message appears urgent and includes a shortened web link.

Current Challenge

He struggles to recognize phishing techniques and finds conventional fact-checking websites difficult to navigate.

How Tanglaw P2P Helps

Using voice-assisted guidance, Roberto checks the message locally.

The Localized Text Parser explains that the message demonstrates common characteristics of phishing, including artificial urgency, impersonation of financial institutions, and suspicious links.

The explanation helps Roberto understand why the message should not be trusted instead of simply telling him to ignore it.

User Persona 3 — Angel

The Student Media Literacy Advocate

Age: 19

Location: State university student

Connectivity: Stable internet on campus, limited connectivity at home

Situation

Angel frequently helps relatives verify forwarded social media posts, political claims, and online scams.

She wants to encourage responsible information sharing within her family and community.

Current Challenge

Existing fact-checking tools require stable internet access and are often too technical for older family members.

How Tanglaw P2P Helps

Angel updates the community Threat Ledger whenever she has campus internet access.

When visiting home, her family benefits from updated offline verification resources.

She also uses the application during community awareness sessions, helping neighbors understand misinformation techniques while strengthening local Media and Information Literacy.

Inclusion and Accessibility

Tanglaw P2P follows a **universal design philosophy**, ensuring that Media and Information Literacy remains accessible regardless of age, literacy level, language, disability, or connectivity.

The platform incorporates:

- Offline-first functionality
- Low-bandwidth operation
- Voice-assisted interaction
- Simplified language

- High-contrast visual design
- Multilingual support
- Explainable verification
- Mobile-first usability
- Privacy-preserving local processing

Rather than assuming all users possess advanced digital skills, the platform adapts to diverse levels of digital confidence and accessibility needs.

5. Success Metrics and Evaluation

Measuring Meaningful Community Impact

The success of Tanglaw P2P will not be measured solely by downloads or technical performance. Instead, the project will evaluate whether it strengthens **Media and Information Literacy (MIL), community resilience, and equitable access to trustworthy information**.

To achieve this, monitoring will combine technical indicators, user adoption metrics, behavioral outcomes, and community impact measures throughout pilot implementation and future scaling.

Key Performance Indicators (KPIs)

A. Media and Information Literacy Outcomes

These indicators evaluate whether users become more confident and capable of critically evaluating information.

Objective	Indicator	Target
Improve verification behavior	Percentage of users who verify suspicious information before sharing	≥70% during pilot
Increase MIL competency	Improvement in pre- and post-pilot MIL assessment scores	≥30% improvement
Improve confidence	Users reporting confidence in identifying misinformation	≥80% positive responses
Encourage responsible sharing	Reduction in self-reported sharing of unverified information	≥40% reduction

B. Community Impact

The project measures whether communities become more resilient to misinformation and fraud.

Objective	Indicator	Target
Reduce misinformation exposure	Decline in recurring misinformation incidents reported by pilot communities	Increasing trend over pilot period
Improve disaster readiness	Users successfully verifying emergency information before acting	≥75% of participants
Improve scam awareness	Increased recognition of phishing and scam indicators	≥35% improvement
Strengthen community participation	Number of active community contributors and local champions	Increasing throughout pilot

C. Accessibility and Inclusion

Media and Information Literacy should be available to everyone regardless of connectivity or digital confidence.

Objective	Indicator	Target
Offline usability	Percentage of verification tasks completed without internet	≥90%
Accessibility	User satisfaction among seniors and first-time smartphone users	≥80% positive
Inclusive participation	Representation of underserved groups in pilot communities	Balanced participation across demographics
Language accessibility	User comprehension of multilingual guidance	≥85% comprehension

D. Technical Performance

Reliable technology is essential for community trust.

Objective	Indicator	Target
Verification speed	Average offline verification response time	<2 seconds
Synchronization efficiency	Size of update packages	Lightweight (<1 MB typical updates)
Reliability	Successful synchronization rate	≥95%
Privacy	Personal data transmitted during local verification	None by design

Evaluation Approach

Project evaluation will combine:

- Baseline and post-pilot Media and Information Literacy assessments.
- Community surveys.
- Focus group discussions.
- User analytics (privacy-preserving).
- Community partner feedback.
- Participatory evaluation workshops.

This mixed-method approach ensures that project success is measured not only through technical performance but also through meaningful improvements in community information resilience.

6. Technology Approach

Technology Guided by Community Needs

Technology in Tanglaw P2P is intentionally selected to support accessibility, affordability, privacy, and long-term sustainability.

Rather than maximizing technological complexity, the project prioritizes technologies that remain reliable in low-resource environments while supporting Media and Information Literacy.

The platform follows four design principles:

- Offline-first
- Low-bandwidth
- Privacy-preserving
- Open and scalable

System Architecture

The platform consists of five integrated components:

1. Progressive Web Application (PWA)

The application functions across modern smartphones without requiring installation through commercial app stores.

This reduces storage requirements while improving accessibility for users with limited devices.

2. Offline Threat Ledger

Verified scam patterns, misinformation indicators, emergency guidance, and educational resources are securely stored on each device.

This enables verification without continuous internet access.

3. Explainable Verification Engine

Instead of functioning as a black-box classifier, the verification engine explains why information appears trustworthy or suspicious by identifying observable indicators such as:

- impersonation,
- urgency,
- suspicious links,
- inconsistent language,
- known misinformation patterns.

This reinforces Media and Information Literacy rather than replacing human judgment.

4. Hybrid Synchronization

Whenever internet access becomes available, compact updates refresh the local Threat Ledger.

Synchronization prioritizes:

- low bandwidth,
- interrupted connectivity,
- privacy,
- efficient update delivery.

5. Community Knowledge Sharing

Communities can distribute updated verification resources through trusted local networks, enabling information resilience even where internet infrastructure remains limited.

Technology Stack (Proposed)

Component	**Technology**
Frontend	React + Progressive Web App
Backend	Node.js/Typescript
Local Storage	IndexedDB / SQLite (depending on platform)
Synchronization	Lightweight REST API with compressed updates
AI/NLP	Lightweight local language analysis for explainable pattern recognition
Security	End-to-end encrypted synchronization, secure local storage
Deployment	Cloud-hosted synchronization service with offline-first architecture

Note: The technology stack may evolve during implementation based on pilot feedback and technical validation, while preserving the project's core principles of accessibility, privacy, and interoperability.

7. Governance and Ethical Framework

Community-Centered Governance

Trust is essential for any Media and Information Literacy intervention.

Tanglaw P2P therefore adopts a governance model that combines technical safeguards with community participation.

Rather than relying solely on automated systems, governance is shared among community stakeholders, educators, local institutions, and technical administrators.

Guiding Principles

The platform follows five governance principles:

Accuracy

Verification resources are based on trusted sources, documented evidence, and transparent review processes.

Transparency

Users receive explanations rather than unexplained automated decisions.

Risk assessments remain understandable and interpretable.

Privacy

Personal messages are analyzed locally whenever possible.

The platform minimizes personal data collection and follows privacy-by-design principles.

Community Participation

Community members contribute to identifying emerging misinformation trends while final verification remains evidence-based.

Accountability

Verification resources are periodically reviewed and corrected as new information becomes available.

Users can report inaccuracies and appeal incorrect classifications.

Data Governance

To protect users, the platform follows these commitments:

- Collect only information necessary for system improvement.
- Avoid storing personal message content on centralized servers whenever possible.
- Use anonymized and aggregated insights for community trend analysis.
- Comply with the Philippine Data Privacy Act of 2012 and internationally recognized responsible AI principles.

AI Governance

Artificial intelligence is designed to **assist human judgment rather than replace it.**

The platform:

- explains recommendations,
- avoids opaque decision-making,
- encourages verification through trusted sources,
- and clearly communicates uncertainty.

Users remain responsible for final decisions.

8. Risk Assessment and Mitigation

Recognizing Responsible Innovation

Every social innovation project carries risks.

Rather than assuming perfect implementation, Tanglaw P2P identifies potential challenges early and incorporates mitigation strategies throughout development.

Risk	Potential Impact	Mitigation Strategy
Outdated Threat Ledger	Incorrect verification guidance	Regular lightweight synchronization and periodic community updates
False positives or false negatives	Reduced trust in verification results	Explainable outputs, human review, and continuous model refinement
Low community adoption	Limited impact	Co-design with communities, user training, and local champions
Digital literacy barriers	Misinterpretation of verification guidance	Voice assistance, simplified interfaces, multilingual support
Privacy concerns	Reduced user trust	Local processing, minimal data collection, privacy-by-design
Connectivity limitations	Delayed synchronization	Offline-first architecture with asynchronous updates
Misinformation evolves rapidly	Verification resources become outdated	Continuous updates through trusted institutional partnerships
Dependence on institutional partners	Sustainability challenges	Diversified partnerships across government, academia, NGOs, and civil society

Ethical Considerations

Tanglaw P2P is designed to empower communities, not to determine truth on their behalf.

The platform therefore:

- encourages critical thinking instead of blind trust,
- supports evidence-based verification,
- avoids political or ideological bias,
- respects user privacy,
- promotes digital inclusion,
- and reinforces informed decision-making rather than automated censorship.

This approach aligns with UNESCO's vision of Media and Information Literacy as a means of empowering citizens to engage responsibly and independently within today's information ecosystem.

REFERENCES

A Case of Media and Information Literacy (MIL) Advocates... (2017). *ejournals.ph*. https://ejournals.ph/article.php?id=24929

Bangko Sentral ng Pilipinas-related reporting. (2025a, March 25). *BSP: Cyber losses hit P5.82 billion in 2024*. *Philstar*. https://www.philstar.com/the-freeman/cebu-business/2025/03/27/2431410/bsp-cyber-losses-hit-p582-billion-2024

Bangko Sentral ng Pilipinas-related reporting. (2025b, June 10). *BSP reports rise in online transaction complaint resolutions*. *Philippine News Agency*. https://www.pna.gov.ph/articles/1251989

Digital fraud in PHl exceeds global rate for sixth year. (2026, May 19). *Daily Tribune*. https://tribune.net.ph/2026/05/20/digital-fraud-in-phl-exceeds-global-rate-for-sixth-year

OECD. (2019). *The effects of zero rating*. OECD Digital Economy Papers, No. 285. https://www.oecd.org/content/dam/oecd/en/publications/reports/2019/07/the-effects-of-zero-rating_45c41e82/6eefc666-en.pdf

Organisation for Economic Co-operation and Development. (2019). *The effects of zero rating*. https://www.oecd.org/content/dam/oecd/en/publications/reports/2019/07/the-effects-of-zero-rating_45c41e82/6eefc666-en.pdf

Philippine Statistics Authority. (2026). *National Information and Communications Technology Household Survey*. https://psa.gov.ph/statistics/nicths

Rising scam exposure in the Philippines underscores need for cross-sector action. (2025, November 12). *GSMA*. https://www.gsma.com/newsroom/press-release/rising-scam-exposure-in-the-philippines-underscores-need-for-cross-sector-action-war

UNESCO. (2023, April 19). *Youth needed as leaders for universal access to media and information literacy*. https://www.unesco.org/en/articles/youth-needed-leaders-universal-access-media-and-information-literacy

UNESCO. (2026a, July 4). *UNESCO Youth Hackathon 2026*. https://www.unesco.org/en/articles/unesco-youth-hackathon-2026

UNESCO. (2026b, January 28). *Media in Crisis Preparedness and Response*. https://www.unesco.org/en/media-crisis

VERA Files. (2020a, January 16). *Mon Tulfo falsely claims Phivolcs failed to issue Taal eruption warning*. https://verafiles.org/articles/vera-files-fact-check-mon-tulfo-falsely-claims-phivolcs-fail

VERA Files. (2020b, January 23). *Video showing Taal volcano on "Alert Level 5" not true*. https://verafiles.org/articles/vera-files-fact-check-video-showing-taal-volcano-alert-level

VERA Files. (2022, March 16). *Netizens falsely claim overseas eruption photos as Taal volcano*. https://verafiles.org/articles/vera-files-fact-check-netizens-falsely-claim-overseas-erupti

VERA Files. (2025, October 9). *Fact check: Eruption video not from Taal volcano*. https://verafiles.org/articles/fact-check-eruption-video-not-from-taal-volcano

World Bank. (2024). *WB report: PH Internet connectivity lags behind Asean countries*. https://newsbytes.ph/2024/04/30/wb-report-ph-internet-connectivity-lags-behind-asean-countries/